USN | | | | | | | | | | [22TCSDE12]

FIFTH SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

CRYPTOGRAPHY

Time: 3 Hours **Maximum Marks: 100**

Instructions:

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	List four major types of security attacks.	1	CO1	**02**
	b.	What is Block cipher? Describe Substitution-Permutation Network (SPN).	1	CO2	**02**
	c.	Describe Digital Signature.	2	CO3	**02**
	d.	Write a note on Kerberos	1	CO4	**02**
	e.	Define: i) Intruders ii) Virus	1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	List all the security services in the OSI Security Architecture.	1	CO1	**05**
		OR			
	b.	Illustrate the classical encryption techniques.	2	CO1	**05**
3.	**a.**	Describe the design principles of Data Encryption Standards(DES) and its strengths.	L2	CO2	**05**
		OR			
	b.	Explain Differential and Linear Crypt analysis.	L2	CO2	**05**
4.	**a.**	In the Diffie-Hellman Key Exchange, given the public parameters p=23 and g=5, if A's private key is a=6 and B's private key is b=15, compute the shared secret key that both A and B will compute?	L3	CO3	**05**
		OR			
	b.	illustrate the working of Digital Signature Authentication Protocol	L2	CO3	**05**
5.	**a.**	Discuss the key components of public key infrastructure (PKI).	L2	CO4	**05**
		OR			
	b.	Explain the working principle of Pretty Good Privacy (PGP).	L2	CO4	**05**
6.	**a.**	Explain the challenges in implementing the Web Security.	L2	CO5	**05**
		OR			
	b.	Explain step-wise working of SSL.	L2	CO5	**05**
7.	**a.**	Apply the principle of Hash function to ensure the message integrity.	L2	CO5	**05**
		OR			
	b.	Assess the effeteness of Secure Electronic Transaction (SET) in ensuring online transaction.	L4	CO6	**05**

		PART-C	6 X 10=60		
8.	**a.**	Assess the key components of a comprehensive network security model. Elaborate on the essential security controls.	L4	CO1	**10**
		OR			
	b.	Explain the fundamental principles of transposition techniques and stenography. Discuss their limitations and compare their effectiveness with modern cryptographic techniques.	L2	CO1	**10**
9.	**a.**	Analyze the strength of the Advanced Encryption Standard (AES) algorithm.	L4	CO2	**10**
		OR			
	b.	Illustrate the steps followed in triple DES. Describe any 3 block cipher modes of operation.	L2	CO2	**10**
10.	**a.**	Discuss the key steps involved in the RSA algorithm for key generation, encryption, and decryption? Explain the mathematical principles underlying the security of RSA.	L2	CO3	**10**
		OR			
	b.	Describe the key challenges in key management, and how can these challenges be addressed to ensure the security of cryptographic systems?	L2	CO3	**10**
11.	**a.**	Elaborate the role of a Certificate Authority (CA) in the X.509 authentication service?	L2	CO4	**10**
		OR			
	b.	What are the key benefits of using S/MIME for email security? How does S/MIME ensure the confidentiality, integrity, and authenticity of email messages?	L2	CO4	**10**
12.	**a.**	Compare and contrast SSL and TLS. How does TLS address the security vulnerabilities present in SSL?	L3	CO5	**10**
		OR			
	b.	Assess the key design principles that should be considered when implementing a firewall to ensure optimal network security?	L4	CO5	**10**
13.	**a.**	Discuss the key components of a trusted system, and what are the challenges building a trusted system? What is its significance?	L2	CO6	**10**
		OR			
	b.	Evaluate the requirements of message authentication. What techniques can be employed to achieve message authentication?	L5	CO6	**10**